

IEEE formatted paper

by Turnitin Report

Submission date: 21-Jan-2025 07:45PM (UTC+0800)

Submission ID: 2568236724

File name: IEEE_formatted_paper.docx (376.68K)

Word count: 4083

Character count: 24458

Enhancing Milk Yield Forecasting in Dairy Farming using an Interpretable Machine Learning Framework

Srinithi.B
Department of Mathematics
Amrita School Of Physical Sciences
Coimbatore
Amrita Vishwa Vidyapeetham
Tamil Nadu, India
cb.sc.i5das21060@cb.students.amrita.edu

Sruthi Nirmala .S.R
Department of Mathematics
Amrita School Of Physical Sciences
Coimbatore
Amrita Vishwa Vidyapeetham
Tamil Nadu, India
cb.sc.i5das21061@cb.students.amrita.edu

5 Senthil Kumar Thangavel
Department of Computer science and Engineering
Amrita school of Engineering
Amrita Vishwa Vidyapeetham
Tamil Nadu, India
t_senthilkumar@cb.amrita.edu

Somasundaram K
Department of Mathematics
Amrita School of Physical Sciences
Coimbatore
Amrita Vishwa Vidyapeetham
Tamil Nadu, India
s_sundaram@cb.amrita.edu

Abstract— This study employs sophisticated machine learning techniques and explainable AI (XAI) to enhance milk yield predictions and optimizes dairy farming practices. Split k-means clustering, KNN, SVM combined with k-means, and regression techniques such as binomial, polynomial, and logistic regression are all used in this methodology. KNN in conjunction with SVM improves prediction accuracy even further. Explainable AI techniques, such as SHAP, provide clarity by emphasizing feature significance, while diverse representations improve data scrutiny. These methods offer practical insights into improving dairy farming operations, ensuring efficient resource utilization, and increasing milk output.

Keywords— Machine Learning, Explainable AI, Milk Yield Prediction, Optimization, SHAP, Dairy Farming

I. INTRODUCTION

One of the main sectors of the agricultural sector is dairy farming, which is focused on producing milk and other dairy products mostly from cows. Dairy farmers have adopted novel tactics, such as Total Mixed Ration (TMR) systems, to improve milk output and guarantee quality.

A. Systems of Total Mixed Ration (TMR):

TMR systems, which provide a methodical and effective approach to feed composition and management, have completely changed the dairy farming industry. TMR minimizes the possibility of ingredient sorting and separation by combining all feed ingredients into a single, nutritionally balanced mixture, ensuring accurate feeding. TMR systems also make it easier and faster to feed bigger groups of cows, albeit they do require careful thought about feeding management procedures and mixer wagons.

B. Machine Learning for Milk Yield Prediction:

Recent advances in data science and machine learning (ML) offers solution to address these issues. Machine learning may

discover hidden patterns in large datasets, offering farmers with predictive insights to help them distribute resources more efficiently. Explainable AI (XAI) improves this technique by

making the forecasts intelligible, resulting in actionable

insights for farm management. This study focuses on the need for precise, intelligible models to forecast milk output. It aims to link traditional dairy practices with precision farming by merging cutting-edge ML approaches and XAI. This study presents a new paradigm for predictive modelling that ensures better decision-making and sustainability in dairy management.

C. Optimization for Improved Milk Yield:

This study uses optimization approaches to fine-tune the distribution of resources, including feed and supplements, in order to further increase milk yield. The model guarantees the effective utilization of these inputs to attain the intended increase in milk yield by utilizing mathematical optimization. In addition to promoting the best possible resource management, this strategy promotes the objective of sustainable dairy farming methods.

II. LITERATURE SURVEY

A. Dairy Farming System :

The research investigations combine conventional dairy farming techniques with robotic dairy farming systems that are automated and equipped with cutting-edge data collection technologies. Advanced technologies like free traffic control, robotic milking equipment, and RFID technology are used for automated milking procedures and data collecting from nursing dairy cows in Queensland, Australia, in a project that focuses on a machine learning framework for robotic dairy farms [1]. These devices make it possible to continuously monitor the environmental conditions, health indicators, and milking habits of individual cows.

Conventional dairy farming, which is distinguished by labour-intensive manual milking procedures and antiquated data gathering techniques, provides a benchmark for

comparison. This conventional method usually entails keeping manual records and evaluating production and health data for each individual cow less frequently.

The research's incorporation of both robotic and conventional dairy farming systems enables a thorough investigation and use of machine learning strategies to enhance dairy farming operations. Through a comparative analysis of machine learning model performance in these various farming systems, researchers can learn more about how well these strategies work to improve milk yield, composition, and overall farm management efficiency.

B. Deep Learning :

An artificial neural network (ANN) model is presented in this study to forecast the first lactation 305-day milk yield in dairy cattle raised on Karan fries [2]. Shows how the ANN model is better than conventional multiple linear regression models and emphasizes how it can improve management and breeding choices. Samples-out random (10-fold random division, 8 for training, 1 for validation, and 1 for testing) and herd/date-out random (70% herds for training, 10% for tuning, and 20% for testing) were the two cross-validation techniques used. MLR is compared to other feed-forward algorithms versions inside the ANN framework, such as Scaled Conjugate Gradient (SCG), Levenberg-Marquardt (LM), Broyden-Fletcher-Goldfarb-Shanno quasi-Newton algorithm (BFGS), and Bayesian Regularization (BR).

As the best ANN model for forecasting milk yield in Frieswal cows, it is chosen based on its lowest RMSE and highest R² values from the testing dataset. To increase forecast accuracy and applicability in actual dairy farming scenarios, more investigation is necessary into the integration of extra variables and environmental elements. The coefficient of determination (R²), the ratio of the expected to observed standard deviations (SDr), the mean absolute error (MAE), the mean percentage error (MPE), the symmetric mean absolute percentage error (SMAPE), and the root mean squared error (RMSE) were among the evaluation measures. To increase the precision and application of predictions, future studies may look into integrating more variables and environmental factors.

Long Short-Term Memory (LSTM) algorithm for dairy cow daily milk yield prediction that is augmented by genetic algorithms [3]. In comparison to traditional LSTM techniques, the hybrid GA-LSTM algorithm demonstrates better accuracy and stability, indicating its potential to improve farm management practices. Although the training and testing methodology's specifics aren't specified, standard procedures probably entail training and testing data split either chronologically or randomly, followed by evaluation using metrics like correlation coefficients, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). To increase prediction accuracy, more research into the integration of environmental components and more input data may be beneficial to the study.

Models for predicting dairy cow milk yield are examined in two experiments by Zhang et al., 2020 and Sharma et al., 2007, where both use cutting-edge techniques, making use of long short-term memory (LSTM) and artificial neural networks (ANN) [2], [3]. Sophisticated algorithms and ANN framework comparisons are two often utilised techniques. In particular, Sharma et al., 2007 used cross-validation methods

such as herd/date-out random and samples-out random, assessing using R², MAE, MPE, SMAPE, and RMSE metrics. In the meantime, (W. Zhang et al., 2020) [8] use genetic algorithms (GA-LSTM) to enhance LSTM for better performance. Although the training and testing procedures aren't explained in full, they probably entail data splitting and correlation coefficient, MAE, RMSE, and MAPE evaluation. Both stress how crucial it is to incorporate input data and ambient elements for accuracy.

C. Machine Learning :

Using a machine learning framework, (B. Ji et al., 2022) [1] attempts to forecast the daily milk yield, milk composition, and frequency of milking for cows in robotic dairy farms. It makes use of information gathered from nursing dairy cows in Queensland, Australia, over a five-year period. The system uses a variety of data inputs, including productivity and behavior data, temperature data, and information particular to cows. It is possible to forecast milk yield, composition, and frequency of milking by using machine learning models. A rolling window approach a size of 28 days was used to split the training and test data. Data from May 2013 to May 2014 were used as the first training data, and values from June after May 2014 were used as the first test data. XGBOOST model was trained with this training and test data, with monthly dataset updates until the entire dataset's expiration in November 2017. The study highlights the value of customized predictive models for effective dairy farm management and shows good prediction accuracy. R-squared, mean residuals, and error distribution are the metrics use for evaluation of the model. The framework's ability to adjust to abrupt changes or novel factors impacting milk output may be limited by its reliance on past data. Furthermore, additional verification in various agricultural environments is required to guarantee the framework's efficacy.

(M. Salamone et al., 2022) [2] Applied machine learning techniques to build predictive models for milk output in dairy cows. It predicts first test day milk yield by using a huge dataset of individual cows and lactation pairs. Samples-out random (10-fold random division, 8 for training, 1 for validation, and 1 for testing) and herd/date-out random (70% herds for training, 10% for tuning, and 20% for testing) were the two cross-validation scenarios used. The study employed Artificial Neural Network (ANN), Random Forest (RF), and Multiple Linear Regression (MLR) as machine learning models. The study emphasizes how predictive algorithms may be used to track individual health and evaluate performance at the herd level. The models' performance appears promising, but more testing and investigation are required to address issues like the prediction of transition failures and the effect of culling on model performance. Predictive ability, mean absolute error (MAE), symmetric mean absolute percentage error (SMAPE), mean absolute percentage error (MAPE), mean square error (RMSE), mean absolute error (MAE), and observed standard deviations (SDr) were among the evaluation measures. Adding more information, like the contents of milk, could improve the algorithms' ability to forecast future.

(Gabriel Machado Dallago et al., 2019) [4] Used multivariate linear regression, random forest, and artificial neural network models to predict the milk yield on the first test day in dairy cows. Samples-out random (10-fold random division, 8 for training, 1 for validation, and 1 for testing) and herd/date-out random (70% herds for training, 10% for tuning,

and 20% for testing were the two cross-validation scenarios used [2]. The relevance of biological explanations for model parameters and the superiority of the artificial neural network model are emphasized in the study. To increase prediction accuracy, greater research into herd variations and the addition of more dairy herd improvement measures are necessary. Predictive capacity, slope prediction, and root mean square error (RMSE) were among the evaluation metrics. The study recognizes that additional research on herd variations and the addition of other dairy herd performance indicators are necessary to improve accuracy.

Bhosale et al., 2024 estimated milk yield in dairy cattle, this study evaluates the effectiveness of multiple linear regression and feed-forward neuro-computing [4]. It shows that the neuro-computing method works better than conventional linear regression models, suggesting that more sophisticated machine learning approaches may be used to increase prediction accuracy. The lack of detail in the study regarding the features or datasets utilized may restrict our ability to comprehend the variables influencing prediction accuracy.

O'Leary et al., 2022 Assesses different machine learning techniques for predicting milk volume in Ireland [5]. The paper finds that Random Forest and K Neighbours Regression are consistently reliable models for milk yield predictions by examining data from various ML models and feature engineering strategies. The results offer perspectives for those involved in the dairy sector and propose ways to enhance the precision of forecasts. Although the study provides insightful information, more research might examine how deep learning systems and real-time data streams can be integrated to increase prediction accuracy.

In a study by Sharpnack et al., 2023, to forecast metritis episodes in dairy calves, machine learning algorithms and pre-processing techniques for sensor data were used [6]. The accuracy of three machine learning algorithms—Random Forest, k-Nearest Neighbors, and Support Vector Machine—in forecasting metritis events is evaluated, and model performance is optimized by adjusting parameters. The models' capacity to detect metritis episodes based on sensor data is assessed using a variety of metrics, such as sensitivity, specificity, accuracy, and area under the curve. When forecasting metritis episodes in dairy cattle, it is discovered that the Random Forest algorithm performs better than the other algorithms. The results point to possible advancements in the treatment of postpartum illness in dairy cows. To enhance early illness diagnosis and management strategies, new study may investigate the integration of more sensors and advanced machine learning models.

Several research use cross-validation methods and other assessment criteria to estimate milk output and associated parameters in dairy cattle using machine learning techniques. Ji et al., 2022, Salamone et al., 2022, and Dallago et al., 2019 highlight the significance of cross-validation and machine learning models like Random Forest and Artificial Neural Networks (ANN), utilising R-squared and RMSE as assessment metrics [1], [7], [8]. To separate training and testing data, they usually employ a cross-validation technique like samples-out random or herd/date-out random.

On the other hand, feed-forward neuro-computing is more effective than linear regression, according to Bhosale et al., 2024, indicating that more thorough data description is

necessary [4]. Random Forest and K Neighbours Regression, which frequently use comparable data splitting techniques, are found to be consistently accurate by O'Leary et al., 2022 and Sharpnack et al., 2023 compare methods such as Random Forest and Support Vector Machine with the goal of forecasting metritis episodes in dairy calves [5], [6]. They propose the integration of additional sensors and sophisticated machine learning models to improve early illness diagnosis and management approaches.

III. METHODOLOGY

In order to predict and maximize milk yield in dairy farms, this study combines Explainable AI (XAI) with machine learning and optimization techniques. To guarantee thorough analysis and decision assistance, the system is divided into three primary stages: evaluation and feature impact analysis, milk yield prediction, and milk yield optimization. Every phase makes use of instruments and methods.

A. Phase I : Milk Yield Prediction

Constructing predictive models that use a variety of input characteristics to estimate milk yield. These characteristics include cow characteristics and feed composition. Establishing a strong and precise predictive framework that can anticipate milk yield and offer comprehensible insights into the connections between input variables and milk production is the goal. To guarantee the quality and consistency of inputs, the data goes through pre-processing procedures such handling missing values, normalization, and feature selection.

1) Model Training and selection:

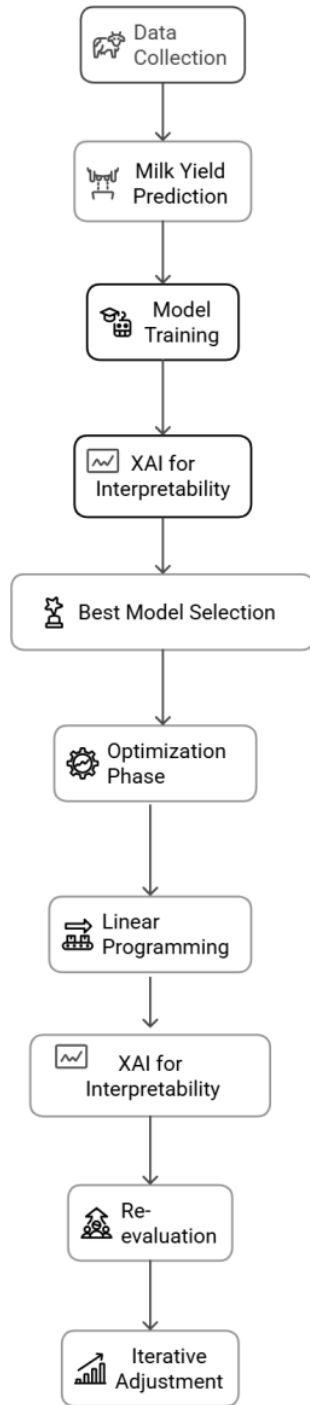
To capture the various patterns and correlations in the dataset, a variety of machine learning models were investigated. These include Random Forest, XGBoost, k-Nearest Neighbors (KNN), Linear Regression, Polynomial Regression, Ridge Regression, Lasso Regression, and Support Vector Machine (SVM) Regression. This phase also included the integration of ensemble techniques to improve prediction accuracy. Predictions from several models were combined using techniques including weighted ensembling, stacking, and blending. Combinations such as SVM Regression + KNN Regression and SVM Regression + Lasso Regression + Ridge Regression, were used to capitalize on the advantages of each model and produce predictions that were more reliable.

2) Explainable AI for Model Interpretability:

After the models were trained, Explainable AI (XAI) approaches were applied to ensure that the framework provided interpretable insights so that stakeholders could act on the predictions. This was done after the best-performing model was identified.

SHAP (SHapley Additive Explanations) was used to analyze the general behavior of the selected model, highlighting the contributions of each feature to the predictions. SHAP provided insights into the factors influencing milk production globally and identified the most significant input features.

Using LIME (Local Interpretable Model-agnostic Explanations), which offered localized insights into the influence of certain input factors on the outcome, individual predictions were explained.



Following the use of these XAI techniques, the selected model was then tested on new prediction scenarios.

B. Phase II : Milk Yield Optimization

Once the top-performing prediction model has been determined, we concentrate on creating an effective feeding plan to maximize milk yield. Maximizing milk production while taking into account realistic limitations such feed availability and dietary needs is the aim. In order to produce outcomes that are both actionable and long-lasting, this phase integrates optimization approaches with the machine learning model's predictive skills.

In this procedure, the information obtained via Explainable AI (XAI) tools like SHAP and LIME is crucial. The optimization framework can prioritize particular feed ingredients or cow management techniques by determining the most important characteristics influencing milk yield.

1) Feed optimization using linear programming:

The optimal distribution of feed ingredients (such as concentrate feed, dry feed, and green fodder) while staying under practical limitations like nutritional limits is determined using linear programming, or LP. The feed ingredients that have the most effects on milk production are prioritized using information from SHAP and LIME.

2) Re-optimization and Iterative Adjustment:

Following optimization, the chosen machine learning model is used to reevaluate the anticipated milk yield.

Once more, the updated predictions are analysed using XAI techniques (SHAP and LIME) to recommend feature changes for even greater milk output improvement.

IV. RESULTS

This section summarizes the outcomes from both phases of the suggested methodology: Milk Yield Optimization (Phase II) and Milk Yield Prediction (Phase I). With the help of tables and figures for clarity, the effectiveness of the prediction models and insights obtained via Explainable AI (XAI) methodologies are examined.

A. Phase I : Milk Yield Prediction

Assessment of Model Performance

Using metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R2 score, this study examined how well many machine learning models performed in forecasting milk yield. K-22rest Neighbors (KNN), Random Forest, XGBoost, Ridge Regression, Lasso Regression, Linear Regression, Polynomial Regression, and Support Vector Machines (SVM) are among the models that were put to the test. To improve prediction accuracy, ensemble methods such 16ighted averaging, stacking, and blending were also used. With an R2 score of 0.918, an RMSE of 0.334, and an MAE of 0.495, Lasso Regression was the most successful model among the others. This suggests that Lasso offers the optimal trade-off between variance explanation and error minimization. Despite their competitive performance (MAE: 0.488, RMSE: 0.571, R2: 0.911), ensemble models like Blended (SVR + Lasso + Ridge) did not significantly outperform Lasso Regression due to their higher complexity. Table 1 provides a summary of the specific metrics for each hyperparameter-tuned model.

TABLE I. RESULTS FOR VARIOUS MODELS:

MODEL	MAE	RMSE	R ² SCORE
Linear Regression	0.495115978	0.333943050	0.917861510
Polynomial Regression	0.508785586	0.357597538	0.912043320
Ridge Regression	0.496899803	0.336919066	0.917129512
Lasso Regression	0.495115978	0.333943050	0.917861510
Support Vector Regression	0.497186244	0.337992707	0.916865434
Random Forest Regression	0.556961682	0.451122697	0.889039352
k-Nearest Neighbour	0.760678211	0.890865599	0.780877742
XGBoost Regressor	0.508200107	0.366167533	0.909935397
Blending (SVR+KNN)	0.5000	0.5846	0.9159
Stacking (SVR+KNN)	0.4912	0.5728	0.9102
Blending (SVR+Lasso+Ridge)	0.4883	0.5714	0.9107
Ensemble Averaging	0.4975	0.5884	0.9148
Weighted Ensemble	0.4965	0.5864	0.9154

B. Explainable AI (XAI) results

Deeper understanding of the impact of important variables on milk yield estimates was made possible by the use of Explainable AI approaches, such as SHAP (SHapley Additive ExPlanations) and LIME (Local Interpretable Model-agnostic Explanations). These techniques confirmed how crucial it is to improve feed composition and management techniques in order to increase milk production.

C. SHAP and LIME Results

According to the SHAP analysis, "Green Fodder (Kg)" had the greatest influence and greatly increased milk output, followed by "Dry Fodder (Kg)" and "Concentrate (Kg)." "Green Fodder (Kg)" continuously showed a positive association with milk yield, confirming its crucial importance, according to the SHAP summary plot. On the other hand, "Azolla (Kg)," although making a good contribution, showed a comparatively smaller effect on predictions, indicating that it ought to be employed as an additional enhancer rather than a fundamental factor.

Further revealing feature dependencies were the SHAP interaction values, which demonstrated that the impact of "Dry Fodder (Kg)" was enhanced when paired with greater values of "Concentrate (Kg)." This emphasizes how crucial it is to balance feed ingredients in order to get the best results.

LIME clarified particular predictions, offering supplementary, localized insights. A 10% increase in "Green Fodder" resulted in a 12% increase in milk yield in one test case, for example, while the "Lactation Period" was crucial in affecting projections for cows with prolonged lactation cycles. Additionally, LIME visualizations showed that raising "Concentrate (Kg)" had a more noticeable impact than other attributes for cows with lower baseline yields, indicating that these situations would benefit from targeted interventions.

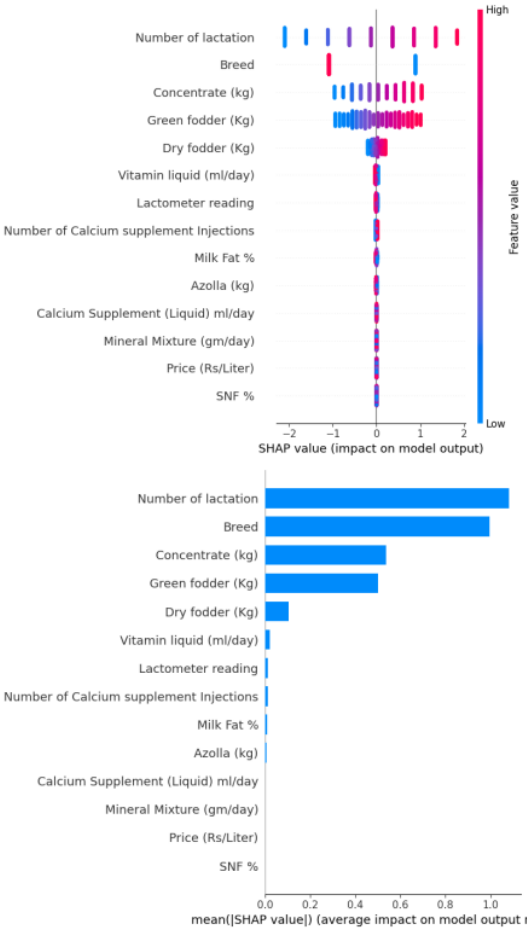


Fig. 1.

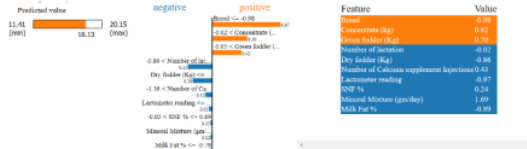


Fig. 2.

Table 2 provides a summary of the combined SHAP and LIME results, while Figures 1 and 2 provide visual insights.

TABLE II. TABLE TYPE STYLES

Feature	SHAP Value Impact	Contribution of LIME (%)	Ranking of Importance
Green Fodder (Kg)	High	45%	1
Dry Fodder (Kg)	Moderate	40%	2
Concentrate (Kg)	Moderate	35%	3

Azolla (Kg)	Low	10%	4
Lactation Period	High	50%	5

TABLE III. RESULTS OF FEED OPTIMIZATION

Feature	Suggested Modification	Change in SHAP Value	Effect on Production
Green Fodder (Kg)	Increase by 10%	+0.15	Significant
Dry Fodder (Kg)	Decrease by 5%	-0.08	Moderate
Azolla (Kg)	Increase by 15%	+0.10	Low

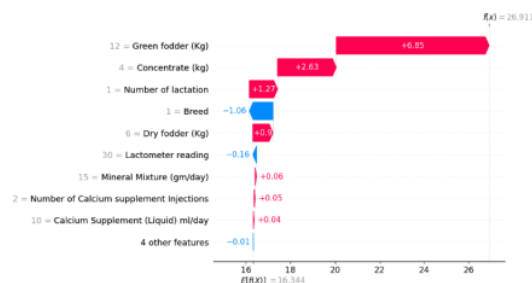


Fig. 3.



Fig. 4.

D. Phase II : Milk Yield Optimization

Optimization Outcomes

In order to optimize milk yield, particular changes to feed mix and management techniques were examined during the optimization phase. Key characteristics that were determined to be crucial in Phase I were the focus of the evaluation. The efficacy of these modifications was confirmed using the performance indicators.

These adjustments were informed by the SHAP analysis, which highlighted Green Fodder as the most impactful feature. Increasing its proportion led to substantial gains in milk yield. To enhance the effectiveness of these adjustments, iterative optimization was employed. This involved cyclically evaluating milk yield predictions using updated feed compositions. After each iteration, SHAP and LIME analyses were conducted to refine the recommendations further. This iterative approach ensured that the feed composition remained balanced while accommodating practical constraints, such as resource availability and nutritional requirements. Additionally, environmental factors, such as seasonal variations and changes in cow lactation cycles, were incorporated into the optimization framework. This adaptive strategy allowed the model to provide dynamic recommendations, making the results more robust and applicable across diverse dairy farm conditions.

The SHAP results were further supported by LIME analysis, which offered case-specific information for individual cows. By using these insights, the optimization tactics were improved and individualized feed recommendations were made. For instance, Figure 4 illustrates how raising "Azolla (Kg)" and "Green Fodder (Kg)" for particular situations resulted in quantifiable production improvements. Depending on the baseline diet and cow characteristics, the expected gains ranged from 5% to 12%. LIME emphasized the trade-offs associated with feed modifications, such as the diminishing rewards of raising "Dry Fodder (Kg)" above a particular level. These realizations promoted sustainable dairy farming methods by lowering feed costs while guaranteeing the intended output increases.

ACKNOWLEDGMENT

'The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.'

REFERENCES

- [1] B. Ji, T. Banhazi, C. J. C. Phillips, C. Wang, and B. Li, "A machine learning framework to predict the next month's daily milk yield, milk composition and milking frequency for cows in a robotic dairy farm," *Biosyst Eng*, vol. 216, pp. 186–197, Apr. 2022, doi: 10.1016/j.biosystemseng.2022.02.013.
- [2] A. K. Sharma, R. K. Sharma, and H. S. Kasana, "Prediction of first lactation 305-day milk yield in Karan Fries dairy cattle using ANN modeling," *Appl Soft Comput*, vol. 7, no. 3, pp. 1112–1120, Jun. 2007, doi: 10.1016/j.asoc.2006.07.002.
- [3] W. Zhang, K. Yang, N. Yu, T. Cheng, and J. Liu, "Daily milk yield prediction of dairy cows based on the GA-LSTM algorithm," in *2020 15th IEEE International Conference on Signal Processing (ICSP)*, IEEE, Dec. 2020, pp. 664–668. doi: 10.1109/ICSP48669.2020.9320926.
- [4] M. D. Bhosale and T. P. Singh, "Comparative study of feed-forward neuro-computing with multiple linear regression model for milk yield prediction in dairy cattle," *JSTOR*.
- [5] C. O'Leary and C. Lynch, "An Evaluation of Machine Learning Approaches for Milk Volume Prediction in Ireland," in *2022 33rd Irish Signals and Systems Conference (ISSC)*, IEEE, Jun. 2022, pp. 1–8. doi: 10.1109/ISSC55427.2022.9826160.
- [6] G. Vidal, J. Sharpnack, P. Pinedo, I. C. Tsai, A. R. Lee, and B. Martínez-López, "Impact of sensor data pre-processing strategies and selection of machine learning algorithm on the prediction of metritis events in dairy cattle," *Prev Vet Med*, vol. 215, p. 105903, Jun. 2023, doi: 10.1016/j.prevetmed.2023.105903.
- [7] M. Salamone *et al.*, "Prediction of first test day milk yield using historical records in dairy cows," *animal*, vol. 16, no. 11, p. 100658, Nov. 2022, doi: 10.1016/j.animal.2022.100658.
- [8] G. M. Dallago *et al.*, "Predicting first test day milk yield of dairy heifers," *Comput Electron Agric*, vol. 166, p. 105032, Nov. 2019, doi: 10.1016/j.compag.2019.105032.

IEEE formatted paper

ORIGINALITY REPORT

9%

SIMILARITY INDEX

8%

INTERNET SOURCES

7%

PUBLICATIONS

4%

STUDENT PAPERS

PRIMARY SOURCES

1

fastercapital.com

Internet Source

1%

2

discovery.researcher.life

Internet Source

1%

3

s3-us-west-2.amazonaws.com

Internet Source

1%

4

export.arxiv.org

Internet Source

1%

5

G. Jeyakumar. "Analyzing the Explorative Power of Differential Evolution Variants on Different Classes of Problems", Lecture Notes in Computer Science, 2010

Publication

<1%

6

academic-accelerator.com

Internet Source

<1%

7

assets-eu.researchsquare.com

Internet Source

<1%

8

fepbl.com

Internet Source

<1%

9	Christian O'Leary, Conor Lynch. "An Evaluation of Machine Learning Approaches for Milk Volume Prediction in Ireland", 2022 33rd Irish Signals and Systems Conference (ISSC), 2022 Publication	<1 %
10	www.currentscience.ac.in Internet Source	<1 %
11	annals-csis.org Internet Source	<1 %
12	veterinaria.org Internet Source	<1 %
13	Boyu Ji, Thomas Banhazi, Clive J.C. Phillips, Chaoyuan Wang, Baoming Li. "A machine learning framework to predict the next month's daily milk yield, milk composition and milking frequency for cows in a robotic dairy farm", Biosystems Engineering, 2022 Publication	<1 %
14	ideas.repec.org Internet Source	<1 %
15	repository.tudelft.nl Internet Source	<1 %
16	www.science.gov Internet Source	<1 %

17	Sharma, A.K.. "Prediction of first lactation 305-day milk yield in Karan Fries dairy cattle using ANN modeling", Applied Soft Computing Journal, 200706 Publication	<1 %
18	acta.imeko.org Internet Source	<1 %
19	tdr.lib.ntu.edu.tw Internet Source	<1 %
20	www.ijsr.net Internet Source	<1 %
21	Submitted to University of Birmingham Student Paper	<1 %
22	naac.iem.edu.in Internet Source	<1 %
23	Wenli Zhang, Kun Yang, Naigong Yu, Tingting Cheng, Jichao Liu. "Daily milk yield prediction of dairy cows based on the GA-LSTM algorithm", 2020 15th IEEE International Conference on Signal Processing (ICSP), 2020 Publication	<1 %
24	coek.info Internet Source	<1 %
25	ouci.dntb.gov.ua Internet Source	<1 %

27 Gabriel Machado Dallago, Darcilene Maria de Figueiredo, Paulo César de Resende Andrade, Roseli Aparecida dos Santos et al. "Predicting first test day milk yield of dairy heifers", Computers and Electronics in Agriculture, 2019

Publication

<1 %

28 M.Rishabha Priya, V. Anandan, V. Anandan, H. Mary Henrietta, S. Varalakshmi, S. Anand. "Recent Innovations in Sciences and Humanities - Highlights the state-of-art developments and innovations which impacts science and engineering.", CRC Press, 2025

Publication

<1 %

29 Shweta Mishra, Avneesh Kumar Singh, Pankaj Prajapati. "Challenges and Opportunities for Innovation in India - Proceedings of International Conference on Challenges and Opportunities in Innovation in India (COII-2024)", CRC Press, 2025

Publication

<1 %